Clinical v.s. Genomic Predictors of Overall Survival in NSCLC

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1. Introduction

Non-small cell lung cancer (NSCLC) accounts for approximately 85% of all lung cancer cases and remains one of the leading causes of cancer-related mortality worldwide. Accurate prediction of survival outcomes for patients with NSCLC is a critical task in both clinical research and personalized medicine. Prognostic models can assist clinicians in estimating patient prognosis and guiding treatment decisions.

Traditionally, clinical variables such as age, sex, cancer stage, and smoking history have been used to model survival outcomes. Among these, tumor stage is widely recognized as one of the strongest prognostic factors. However, advances in genomic sequencing technologies have made large-scale genomic data increasingly accessible, raising the possibility that genomic features could enhance survival predictions beyond traditional clinical variables alone.

Recent studies suggest that genomic alterations may influence tumor progression and treatment response in lung cancer. The Fraction Genome Altered (FGA) reflects the burden of chromosomal instability, a hallmark of aggressive tumors. Tumor Mutational Burden (TMB) quantifies the total number of mutations per megabase and is associated with neoantigen load and immunotherapy response. Incorporating these genomic variables into survival models may provide additional predictive information beyond what clinical factors alone can offer.

The goal of this study is to evaluate whether genomic features can enhance the prediction of overall survival (OS) in patients with NSCLC. Specifically, three survival models are compared: (1) a Cox proportional hazards model using clinical variables only; (2) a Cox model including both clinical and genomic variables; and (3) a Random Survival Forest (RSF) model. Model performance is assessed using 5-fold cross-validated concordance index (C-index), and patients are stratified into clinically meaningful risk groups based on RSF-predicted risk scores. I hypothesize that incorporating genomic features will improve the predictive performance of survival models compared to using clinical variables alone. Additionally, I expect that machine learning approaches, such as Random Survival Forest, may outperform traditional Cox models by capturing non-linear relationships and interactions among predictors.

2. Variables and Data Cleaning

The dataset used in this study was obtained from the MSK-CHORD study, a large-scale genomic dataset published by the Memorial Sloan Kettering Cancer Center in 2024 [1]. This dataset includes clinical and genomic information for patients diagnosed with non-small cell lung cancer. The primary outcome variable is overall survival (OS), defined as the time from diagnosis to

death or last follow-up. The event indicator was coded as 1 for death and 0 for censored observations (patients alive at last follow-up). No progression information was used in this analysis. An event indicator was coded as 1 if the patient was deceased (DECEASED) and 0 if censored (LIVING) at the last follow-up.

Several clinical variables were incorporated into the analysis: age at diagnosis (continuous), sex (Female / Male), cancer stage (Stage 1–3 vs. Stage 4), and smoking history (Never / Former or Current Smoker / Unknown). Three genomic features were included: Fraction Genome Altered (FGA), Mutation Count, and Tumor Mutational Burden (TMB, nonsynonymous).

Prior to the analysis, standard data cleaning procedures were performed. These included removing observations with missing survival time, converting categorical variables to appropriate factor levels, and ensuring numeric variables were properly formatted. Observations with missing values in any genomic variable were excluded from the genomic analyses. The final cleaned dataset included n = 7,429 patients with complete data on all clinical and genomic variables, out of an original N = 7,809 patients. A total of 3,820 events (51.4%) were observed during the follow-up period.

3. Statistical Methods

Several statistical methods were used to analyze survival patterns and evaluate predictive performance.

3.1 Kaplan-Meier Survival Analysis

Kaplan-Meier (KM) survival curves were used to describe overall survival patterns in the cohort [2]. The KM estimator is a non-parametric method used to estimate survival probabilities over time. An overall survival curve was generated to characterize the general survival experience of the cohort. Additionally, KM curves were stratified by cancer stage and by RSF-predicted risk groups to examine differences in survival outcomes across subgroups. The log-rank test was used to assess the statistical significance of differences between groups.

3.2 Cox Proportional Hazards Models

Cox proportional hazards (PH) models were used to evaluate the association between covariates and the hazard of death [3]. The Cox model is a semi-parametric survival model that estimates hazard ratios while making no assumptions about the baseline hazard shape. Two Cox models were fitted: Model 1 included clinical variables only (age, sex, stage, smoking history), and Model 2 additionally included genomic variables (FGA, Mutation Count, TMB). The proportional hazards assumption was assessed using Schoenfeld residuals.

3.3 Random Survival Forest

A Random Survival Forest (RSF) model was implemented as a machine learning alternative to Cox regression [4]. RSF extends the random forest algorithm to right-censored survival data by constructing an ensemble of survival trees and averaging their predicted survival functions. This

approach can capture non-linear relationships, high-order interactions, and complex variable effects without assuming proportional hazards. The model was grown with ntree = 500 trees and variable importance was calculated using the permutation-based method.

3.4 Model Evaluation and Risk Stratification

Model performance was evaluated using the concordance index (C-index), which measures the ability of a model to correctly rank survival times based on predicted risk scores. The C-index was chosen as it is a standard metric for evaluating discrimination ability in survival models, reflecting how well the model ranks patients by risk. A C-index of 0.5 indicates performance equivalent to chance, while 1.0 indicates perfect discrimination. To obtain reliable and generalizable performance estimates, 5-fold cross-validation was implemented: the dataset was randomly partitioned into 5 equal-sized folds, and the models were trained on 4 folds and validated on the remaining fold in each iteration. The mean and standard deviation of the C-index across folds were reported. A random seed (set.seed(629)) was used to ensure reproducibility.

For risk stratification, predicted risk scores from the RSF model were used to classify patients into three groups — Low Risk, Intermediate Risk, and High Risk — based on the lower and upper tertiles (33rd and 67th percentiles) of the predicted risk distribution.

4. Results

4.1 Descriptive Statistics

The final analysis included n = 7,429 patients with NSCLC and complete data on all clinical and genomic variables. The median age at diagnosis was 71 years (range: 10–89). The cohort was slightly more female (58.4%) than male (41.6%). The majority of patients had Stage 1–3 disease (56.2%), with the remaining 43.8% diagnosed at Stage 4. Regarding smoking history, 70.3% were former or current smokers, 25.3% were never smokers, and 4.4% had unknown smoking status. Genomic characteristics included a median FGA of 0.099 (range: 0–1.0), median Mutation Count of 6 (range: 1–172), and median TMB of 5.74 mutations/Mb (range: 0–167.4). A total of 3,820 deaths (51.4%) were observed during the follow-up period.

4.2 Survival Patterns

Kaplan-Meier survival analysis revealed a gradual decline in survival probability over time for the overall cohort, with a median OS of 22.2 months. When survival curves were stratified by cancer stage, clear and substantial differences were observed between Stage 1–3 and Stage 4 groups (Figure 1). Patients with early-stage disease demonstrated substantially better survival outcomes compared with those diagnosed at Stage 4. The log-rank test confirmed that these differences were statistically significant ($p < 0.001$).

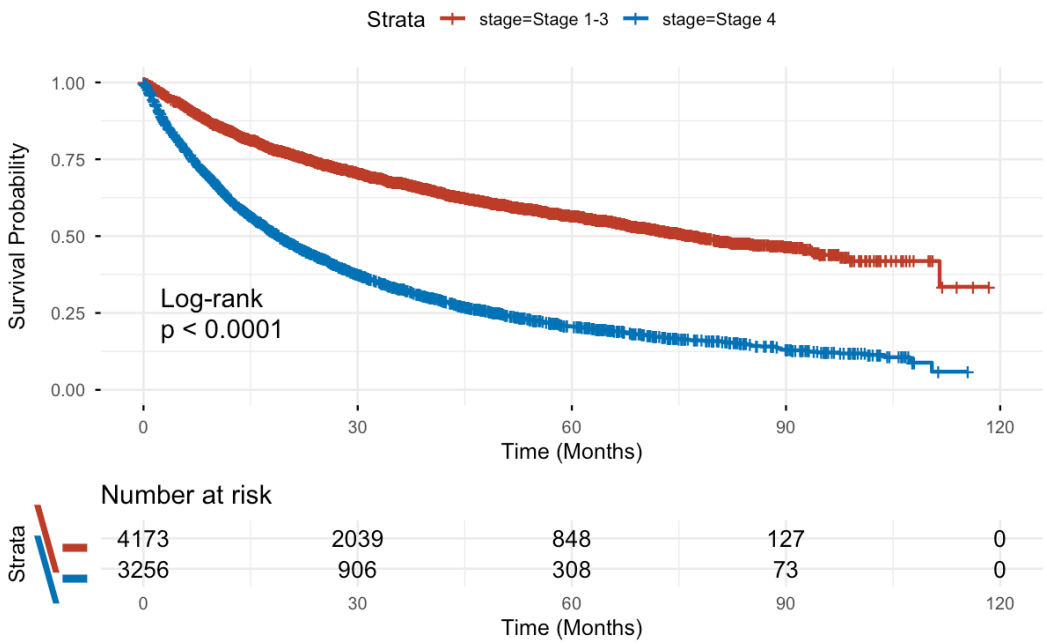

Figure 1. Kaplan-Meier curves of overall survival stratified by cancer stage (Stage 1–3 vs. Stage 4). The log-rank test indicated statistically significant differences between stage groups ($p < 0.001$).

4.3 Model Performance

Three predictive models were constructed and evaluated using 5-fold cross-validation. Table 1 summarizes the cross-validated C-index for each model.

Model	Mean C-index	SD
Clinical Cox (age, sex, stage, smoking)	0.668	0.015
Clinical + Genomic Cox (above + FGA, Mutation Count, TMB)	0.678	0.015
Random Survival Forest (500 trees)	0.681	0.014

Table 1. Five-fold cross-validated concordance index (C-index) for three survival models.

As shown in Table 1, the Clinical Cox model achieved a mean C-index of 0.668 (SD = 0.015). Adding genomic features (FGA, Mutation Count, TMB) to the Cox model increased the C-index to 0.678 (SD = 0.015), representing a modest but consistent improvement of +0.010 over the clinical-only model. The Random Survival Forest model achieved the highest mean C-index of 0.681 (SD = 0.014), outperforming both Cox models. The small standard deviations across folds (0.014–0.015) indicate stable and consistent model performance. These results suggest that genomic features provide additional predictive value, although the improvement is modest, and that machine learning approaches may further enhance model performance.

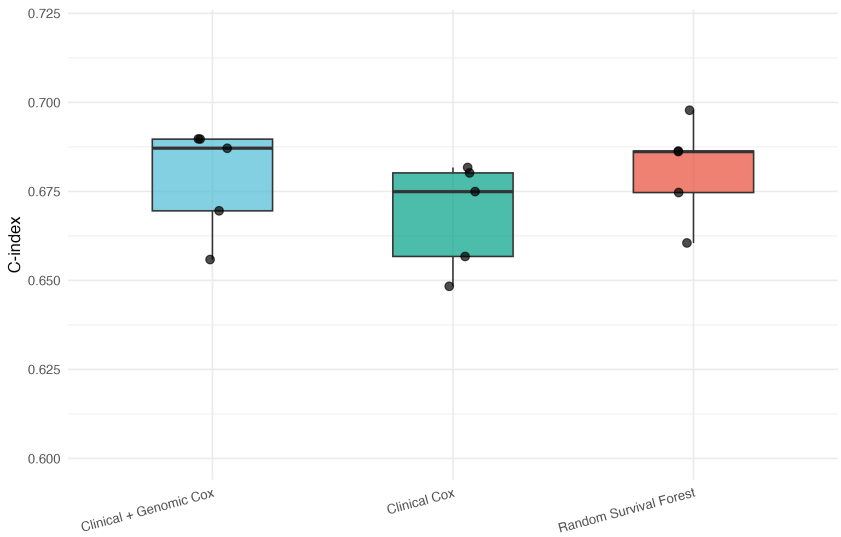

Figure 2. Comparison of 5-fold cross-validated concordance index (C-index) across three survival models. Error bars represent the standard deviation across folds.

4.4 Variable Importance

Variable importance measures from the RSF model were used to identify the most influential predictors of overall survival. As shown in Figure 3, cancer stage was the single most important predictor (importance = 0.205), followed by Fraction Genome Altered (importance = 0.099) and Mutation Count (importance = 0.053). Tumor Mutational Burden and age showed moderate importance (0.043 each), while smoking history and sex contributed less. Notably, two of the top three predictors were genomic variables, highlighting the added value of genomic features in this prediction context.

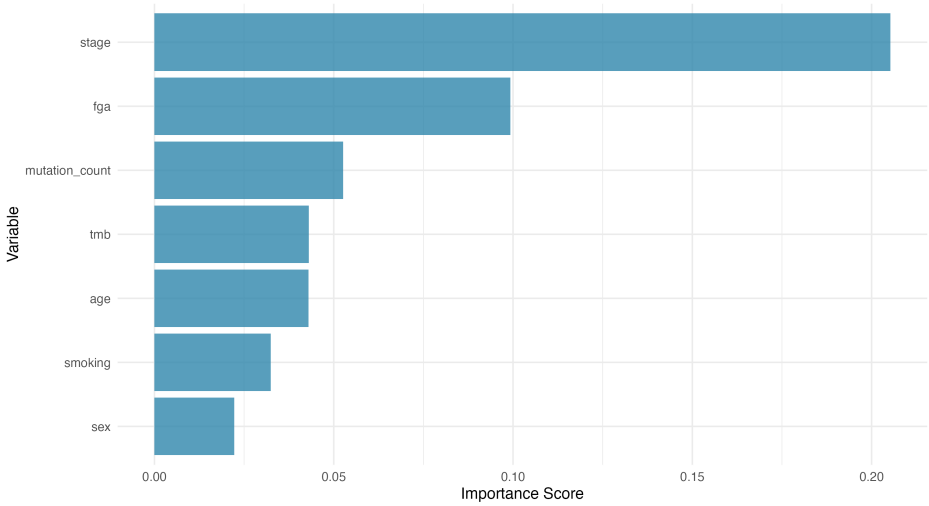

Figure 3. Variable importance from the Random Survival Forest model, ranked by importance score. Higher values indicate greater predictive contribution.

4.5 Risk Stratification

Patients were classified into three risk groups based on the tertiles of RSF-predicted risk scores. As shown in Table 2, the risk groups exhibited markedly different survival outcomes. The Low Risk group (n = 2,477) had a median OS of 34.3 months and an event rate of 28.0%, while the High Risk group (n = 2,476) had a median OS of only 13.5 months and an event rate of 73.1%. The Intermediate Risk group (n = 2,476) showed intermediate survival (median OS = 22.2 months, event rate = 53.2%).

Risk Group	N	Median OS (months)	Event Rate (%)
Low Risk	2,477	34.3	28.0
Intermediate Risk	2,476	22.2	53.2
High Risk	2,476	13.5	73.1

Table 2. Survival outcomes by RSF-predicted risk groups (tertiles).

Kaplan-Meier survival curves by risk group confirmed clear separation among all three groups (Figure 4). All pairwise log-rank tests were statistically significant at $p < 0.001$, demonstrating that the RSF model successfully stratifies patients into clinically meaningful prognostic subgroups. This demonstrates the potential clinical utility of the RSF model in identifying patient subgroups with significantly different prognoses.

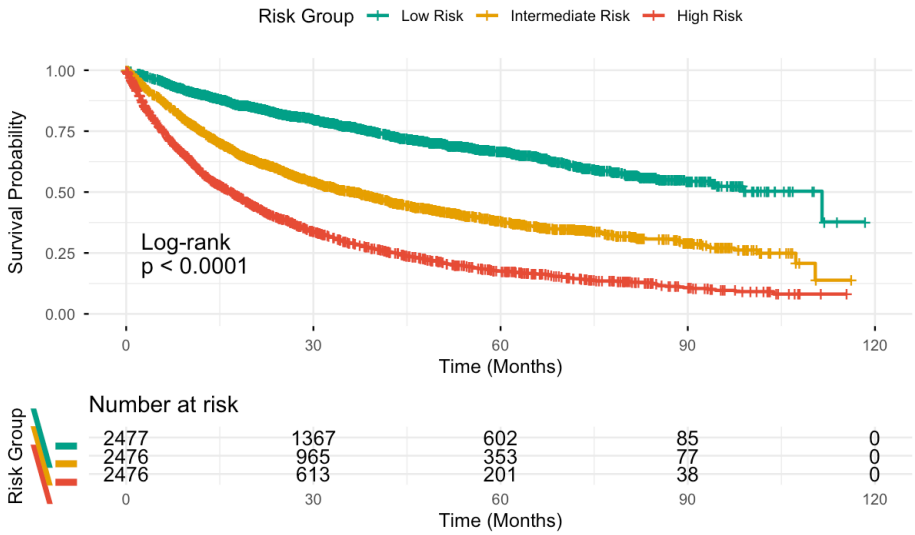

Figure 4. Kaplan-Meier survival curves by RSF-predicted risk groups (Low, Intermediate, High Risk). All pairwise log-rank tests were significant ($p < 0.001$).

5. Discussion

This study investigated whether the integration of genomic features enhances the prediction of overall survival in NSCLC patients beyond what clinical variables alone can provide. Using 5-fold cross-validation, we obtained more reliable and generalizable performance estimates compared with our interim analysis, in which the RSF model was evaluated on the same dataset

used for training (C-index = 0.687). The cross-validated RSF C-index of 0.681 is a more conservative and credible estimate of true predictive performance.

The Random Survival Forest model demonstrated the best predictive performance among the three models evaluated, achieving a cross-validated C-index of 0.681. This suggests that machine learning approaches can capture complex non-linear relationships and high-order interactions among predictors that are not fully captured by traditional Cox regression. The modest but consistent improvement from adding genomic features to the Cox model (+0.010 in C-index) indicates that genomic information provides incremental predictive value beyond clinical variables alone.

The biological relevance of the genomic features is further supported by the variable importance analysis. Fraction Genome Altered ranked second in importance, reflecting the degree of chromosomal instability — a hallmark of aggressive tumor behavior. Mutation Count and TMB also showed meaningful importance, suggesting that tumor mutational burden is a relevant predictor of survival outcomes in this NSCLC cohort.

Several methodological limitations must be acknowledged. First, all models were validated using internal cross-validation only; external validation on independent cohorts is essential to confirm the clinical generalizability of these results. Second, the exclusion of patients with missing genomic data reduced the effective sample size from 7,809 to 7,429, potentially introducing a selection bias toward patients with more complete clinical and genomic records. Third, the analysis focused on a selected subset of genomic features; additional genomic biomarkers (e.g., specific driver mutations, transcriptomic signatures) may further improve predictive performance.

Looking forward, the analysis could be strengthened in several ways. External validation using independent NSCLC datasets — such as The Cancer Genome Atlas (TCGA) or other institutional cohorts — would establish the broader applicability of these models. Incorporating additional biomarker modalities, such as transcriptomic or proteomic profiles, may capture complementary aspects of tumor biology. Advanced deep learning survival models (e.g., DeepSurv, Cox-nnet) capable of handling multi-modal data offer promise for further improving predictive accuracy. Ultimately, the development of a clinical risk calculator based on these models could translate the findings into a practical tool for prognostic decision support in NSCLC management.

References

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